Fundamental Trigonometric Identities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

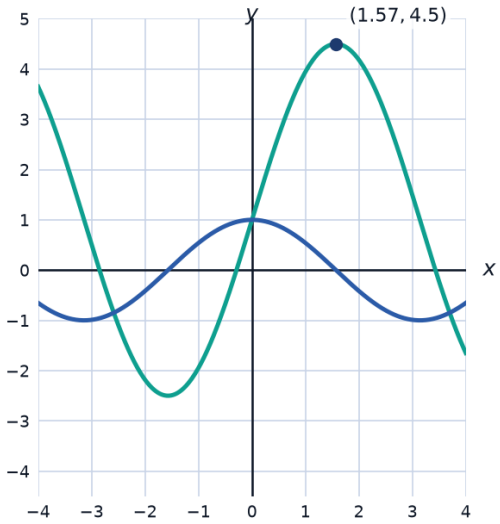
1 [Foundational] • Target: Using the Pythagorean identities

Given that $\sin\theta = \frac{3}{5}$ and θ lies in Quadrant II, find:

a $\cos\theta$ **b** $\tan\theta$ **c** $\sec\theta$ **d** $\csc\theta$

2 [Foundational] • Target: Using the Pythagorean identities

Given that $\tan\theta = 2$ and θ lies in Quadrant III, find $\sin\theta$ and $\cos\theta$ in exact simplified form.



3 [Foundational] • Target: Simplifying and verifying

Simplify each expression to a single trigonometric function:

a $\frac{1-\cos^2 x}{\sin x}$

b $\sec x \cot x$

c $\tan x \csc x$

4 [Foundational] • Target: Simplifying and verifying

Verify the identity $(1+\tan^2 x)\cos^2 x = 1$.

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5 [Foundational] • Target: Quotient and reciprocal identities

Rewrite each expression using only $\sin x$ and $\cos x$, then simplify:

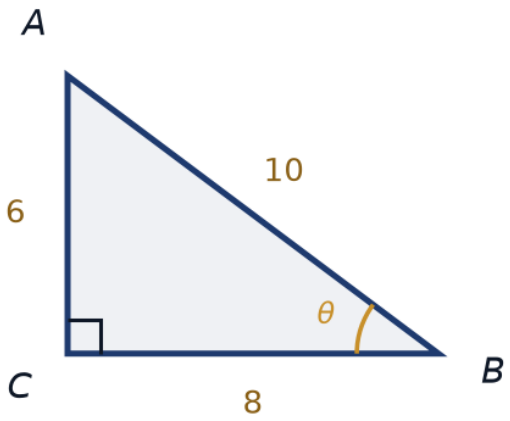
a $\cot x \sin x$

b $\frac{\sec x}{\csc x}$

c $1 - (\sin^2 x \csc^2 x)\sin^2 x$

6 [Foundational] • Target: Quotient and reciprocal identities

Simplify $\frac{\cos x}{1 - \sin x} - \tan x$ to a single term.



7 [Foundational] • Target: More identity verifications

Verify the identity $\sec^2 x - \tan^2 x = 1$ using the Pythagorean identity for tangent and secant.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Identities with sums and products

Simplify $(\sec x - \tan x)(\sec x + \tan x)$ to a constant.

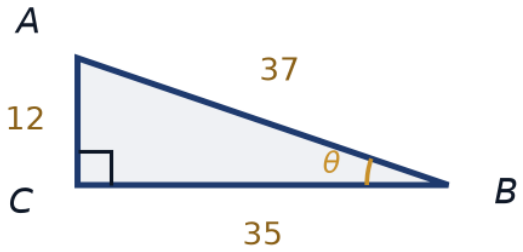
9 [Intermediate] • Target: Identities involving all three Pythagorean forms

State all three forms of the Pythagorean identity (for sine/cosine, tangent/secant, and cotangent/cosecant).

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10 [Intermediate] • Target: Identities involving all three Pythagorean forms

Given $\cot\theta = \frac{5}{12}$ and θ in Quadrant I, find $\csc\theta$ and $\sin\theta$ using the cotangent-cosecant identity.



11 [Intermediate] • Target: Simplifying more complex expressions

Simplify $\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}$ using the sum-of-cubes factoring pattern.

12 [Intermediate] • Target: Numeric identity verification (checking a specific angle)

Verify the identity $1 + \tan^2 x = \sec^2 x$ numerically at $x = \frac{\pi}{6}$ by computing both sides.

13 [Intermediate] • Target: Identities with squared expressions

Given that $\sec\theta = -\frac{13}{5}$ and θ lies in Quadrant II, use the Pythagorean identity $1 + \tan^2\theta = \sec^2\theta$ to find $\tan\theta$ exactly, and then find $\sin\theta$ using the quotient identity.

14 [Intermediate] • Target: Identities with squared expressions

Verify the identity $\frac{\csc x - 1}{\cot x} = \frac{\cot x}{\csc x + 1}$ by cross-multiplying and using the Pythagorean identity.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

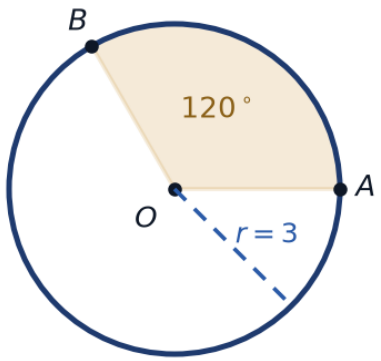
15 [Challenging] • Target: Identities with sums and products • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

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16 [Challenging] • Target: Identities involving all three Pythagorean forms • Cross-Lesson: 8.02 Graphing Rational Functions

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



17 [Challenging] • Target: Simplifying more complex expressions • Cross-Lesson: 8.03 Introduction to Limits

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

18 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.

19 [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 9.02 Geometric Sequences and Series

Solve the logarithmic equation $\log_2(x + 3) + \log_2(x - 3) = 4$ for all real solutions x. State any extraneous roots.

20 [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 9.03 The Binomial Theorem

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Using the Pythagorean identities

Divide $f(x) = 2x^3 + 5x^2 + 5x + 2$ by $(x + 1)$ using synthetic division. Find the quotient.

a. $2x^2 + 3x + 2$

b. $2x^2 + 3x - 2$

c. $2x^2 - 3x + 2$

d. $3x^2 + 3x + 2$

22 [Foundational] • Target: Simplifying and verifying

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

a. 2

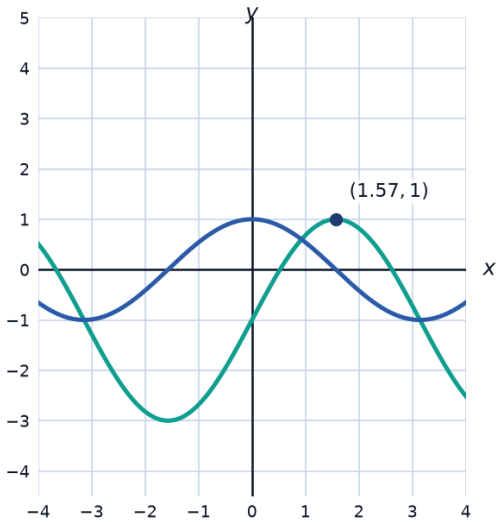
b. 4

c. 8

d. 16

23 [Foundational] • Target: Quotient and reciprocal identities

Consider the exponential growth model described by the function $P(t) = P_0e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.



a. 0.347

b. 0.416

c. 0.301

d. 0.477

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24 [Foundational] • Target: More identity verifications

A complex number z is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

a. $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$

b. $5 + i5$

c. $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

d. $5i$

25 [Foundational] • Target: Identities with sums and products

Find all real roots of $f(x) = x^3 - 4x^2 - 36x + 144$.

a. x=-6

b. x=-6,4

c. x=-5,4,6

d. x=-6,4,6

26 [Foundational] • Target: Identities involving all three Pythagorean forms

Find the common difference of the arithmetic sequence: -1, -2, -3, -4, ...

a. 0

b. -2

c. 1

d. -1

27 [Foundational] • Target: Simplifying more complex expressions

Find the 4th term of the geometric sequence with $a_1 = 2$ and common ratio $r = 2$.

a. 18

b. 17

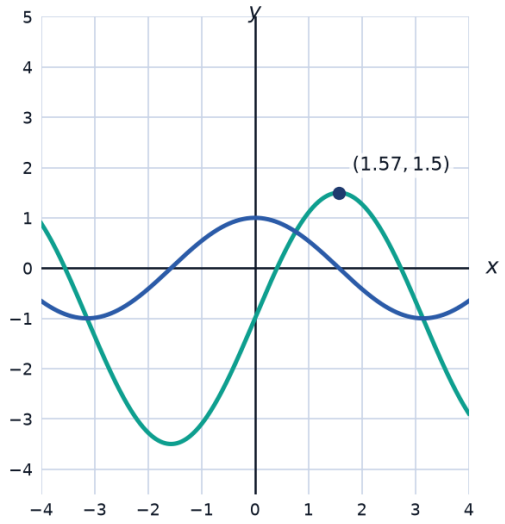
c. 32

d. 16

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28 [Foundational] • Target: Verifying identities with a common denominator strategy

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference d. The terms a_1, a_3, and a_4 form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the n-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?



a. $\frac{5}{4}$

b. $\frac{81}{64}$

c. $\frac{9}{8}$

d. $\frac{5}{8}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Numeric identity verification (checking a specific angle)

Using a double-angle identity, evaluate $\cos^2\theta - \sin^2\theta$ when $\theta = 15^\circ$.

a. -1

b. $-\frac{\sqrt{3}}{2}$

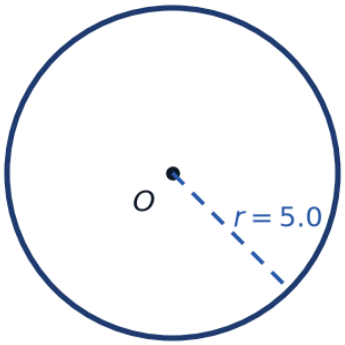
c. $\frac{\sqrt{2}}{2}$

d. $\frac{\sqrt{3}}{2}$

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30 [Intermediate] • Target: Identities with squared expressions

Find the center and radius of $(x+6)^2+(y-1)^2=25$.



a. Center $(6, -1)$, radius 5

b. Center $(-6, 1)$, radius 5

c. Center $(1, -6)$, radius 5

d. Center $(-6, 1)$, radius 25

31 [Intermediate] • Target: Using the Pythagorean identities

Using complex numbers, find $\cos^4\theta$ as a sum of cosines.

a. $\frac{3+4\cos 2\theta+\cos 4\theta}{8}$

b. $\frac{1+\cos 4\theta}{2}$

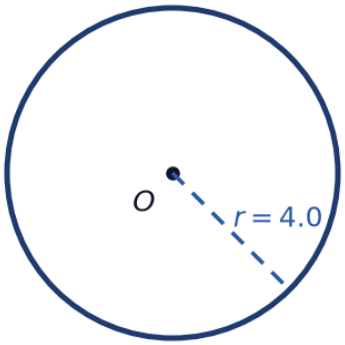
c. $3/8$

d. $\cos 4\theta/8$

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32 **[Intermediate]** • **Target: Simplifying and verifying**

What is the center of the circle $(x-3)^2+(y+2)^2=16$?



a. (-3, 2)

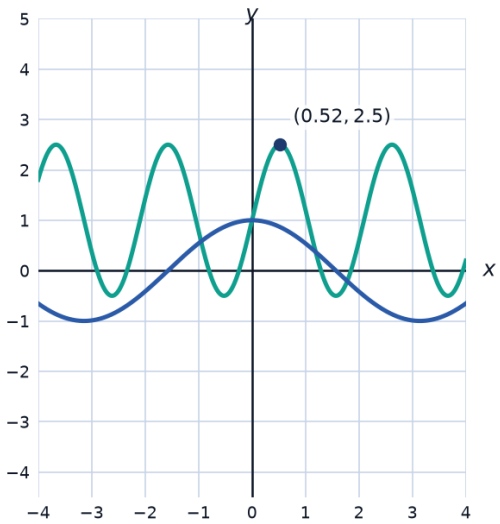
b. (3, -2)

c. (3, 2)

d. (-2, 3)

33 **[Intermediate]** • **Target: Quotient and reciprocal identities**

For $z=re^{i\theta}$, express $\mathrm{Re}(z^3)$ using triple angle formula.



a. r^3

b. $3r^3\cos\theta$

c. $r^3\cos 3\theta$

d. $r\cos 3\theta$

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34 [Intermediate] • Target: More identity verifications

Evaluate $\sum_{k=1}^{n} \cos k\theta$ using complex exponentials.

a. 0

b. $n\cos\theta$

c. $\frac{\sin(n\theta/2)}{\sin(\theta/2)}\cos\frac{(n+1)\theta}{2}$

d. $\cos(n\theta)$

35 [Intermediate] • Target: Identities with sums and products

State $\cos 2\theta$ in three equivalent forms.

a. $\cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$

b. $\cos\theta + \sin\theta$

c. $2\sin\theta\cos\theta$

d. $\cos^2\theta$

36 [Intermediate] • Target: Identities involving all three Pythagorean forms

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of a_{2023}.

a. 2

b. 0

c. -2

d. 1

37 [Intermediate] • Target: Simplifying more complex expressions

A population of bacteria, initially P_0 cells, grows according to the logistic model $P(t) = \frac{L}{1 + Ae^{-kt}}$, where t is in hours, L is the carrying capacity, and A and k are positive constants. At $t = 0$, the population is $P(0) = \frac{L}{3}$. After 2 hours, the population has doubled to $P(2) = \frac{2L}{3}$. Find the exact value of e^{2k} in terms of L and P_0.

a. $\frac{9}{4}$

b. $\frac{4}{9}$

c. 3

d. $\frac{3}{2}$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Consider the exponential function $f(x) = 2^x$. If $f(a) = 8$, what is the value of a?

a. 1

b. 2

c. 3

d. 4

39 [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 9.02 Geometric Sequences and Series

Consider the exponential function $f(x) = 2^x$. What is the value of $f(3)$?

a. 6

b. 8

c. 9

d. 5

40 [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 9.03 The Binomial Theorem

Find the vector projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 2, 0 \rangle$.

a. $\langle 0, 3 \rangle$

b. $\langle 4, 0 \rangle$

c. $\langle 3, 0 \rangle$

d. $\langle 3, 4 \rangle$

41 [Challenging] • Target: Using the Pythagorean identities • Cross-Lesson: 1.01 Functions and Function Notation

For $f(x) = \frac{x^2 - 25}{x - 5}$ when $xeq5$, and $f(5) = k$, find k so that f is continuous at $x = 5$.

a. 10

b. 5

c. 9

d. 11

42 [Challenging] • Target: Simplifying and verifying • Cross-Lesson: 1.02 Domain and Range

The arithmetic sequence with $a_1 = 3$ and $d = 4$ has $a_n = 99$. Find n.

a. 25

b. 99

c. 26

d. 24

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43 [Challenging] • Target: Quotient and reciprocal identities • Cross-Lesson: 1.03 Average Rate of Change

Solve for the larger value of x: $2^{2x} - 10 \cdot 2^{x} + 16 = 0$.

a. 5

b. 3

c. 1

d. 4

44 [Challenging] • Target: More identity verifications • Cross-Lesson: 2.01 Right-Triangle Trigonometry

For the parabola $y^2 = -20x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(-5, 0)$, directrix $x = 5$

b. Vertex $(0, 0)$, focus $(-10, 0)$, directrix $x = 10$

c. Vertex $(0, 0)$, focus $(0, -5)$, directrix $y = 5$

d. Vertex $(-5, 0)$, focus $(0, 0)$, directrix $x = -5$

45 [Challenging] • Target: Identities with sums and products • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the 10th term of the arithmetic sequence with $a_1 = 3$ and $d = 4$.

a. 36

b. 35

c. 39

d. 43

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